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Signal/noise determination apparatus, method, and recording medium

Abstract

A signal/noise determination apparatus includes a plurality of sensors, a determination model recording section, and a signal/noise determining section. The plurality of sensors measure a signal and a noise. The determination model recording section records a determination model used to determine whether components of results of measurement by the sensors expected with hypothetical signal information and hypothetical noise information are from a signal source or a noise source. The determination model is generated by machine learning with the measurement results, the hypothetical signal information, and the hypothetical noise information as training data. The signal/noise determining section determines whether components of the measurement results are from the signal source or the noise source based on the measurement results and the determination model. The signal information includes the position of the signal source and the signal, and the noise information includes the position of the noise source and the noise.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to signal and noise measuring.

Description of the Related Art

(2) There has conventionally been known separating a measured signal into a signal component from a signal source of interest and a signal component from an environmental magnetic noise source (see Japanese Patent Application Publication Nos. 2021-123242, 2009-257933, 2009-113717, 2001-120511, 2012-179352, and Pierpaolo Croce, et al. (5 others), “Deep Convolutional Neural Networks for Feature-

Less Automatic Classification of Independent Components in Multi-Channel Electrophysiological Brain Recordings”, IEEE Transactions on Biomedical Engineering, August, 2019, Volume: 66, Issue: 8, Page(s): 2372-2380, for example). Among others, approaches that use neural networks are known (see Japanese Patent Application Publication Nos. 2009-113717, 2001-120511, 2012-179352, and Pierpaolo Croce, et al. (5 others), “Deep Convolutional Neural Networks for Feature-Less Automatic Classification of Independent Components in Multi-Channel Electrophysiological Brain Recordings”, IEEE Transactions on Biomedical Engineering, August, 2019, Volume: 66, Issue: 8, Page(s): 2372-2380, for example).

SUMMARY OF THE INVENTION

(3) However, the use of a neural network requires a large amount of training data, which is difficult to acquire.

(4) It is hence an object of the present invention to easily acquire training data for machine learning used when it is determined whether components of a measured signal are from a signal source or a noise source.

(5) According to the present invention, a signal/noise determination apparatus, includes: a plurality of sensors arranged to measure a signal and a noise; a determination model recording section that records a determination model used to determine whether components of results of measurement by the sensors expected with hypothetical signal information and hypothetical noise information are from a signal source or a noise source, the determination model generated by machine learning with the measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and a signal/noise determining section arranged to determine whether components of the measurement results are from the signal source or the noise source based on the measurement results and the determination model, wherein the signal information includes the position of the signal source and the signal, and the noise information includes the position of the noise source and the noise.

(6) According to the thus constructed signal/noise determination apparatus, a plurality of sensors measure a signal and a noise. A determination model recording section records a determination model used to determine whether components of results of measurement by the sensors expected with hypothetical signal information and hypothetical noise information are from a signal source or a noise source, the determination model generated by machine learning with the measurement results, the hypothetical signal information, and the hypothetical noise information as training data. A signal/noise determining section determines whether components of the measurement results are from the signal source or the noise source based on the measurement results and the determination model. The signal information includes the position of the signal source and the signal. The noise information includes the position of the noise source and the noise.

(7) According to the signal/noise determination apparatus of the present invention, the signal source may be assumed to be arranged within a predetermined single region, and the noise source may be assumed to be arranged around the region.

(8) According to the signal/noise determination apparatus of the present invention, the signal source or the noise source may be assumed to be arranged randomly.

(9) According to the signal/noise determination apparatus of the present invention, the components may be obtained through independent component analysis on the measurement results.

(10) According to the signal/noise determination apparatus of the present invention, the components may be obtained through principal component analysis on the measurement results.

(11) According to the signal/noise determination apparatus of the present invention, the components may be obtained through FFT on the measurement results and further through IFFT on the result of the FFT.

(12) According to the signal/noise determination apparatus of the present invention, the machine learning may be based on a convolutional neural network.

(13) According to the signal/noise determination apparatus of the present invention, a sigmoid function may be used as an activation function for an output layer of the convolutional neural network.

(14) According to the signal/noise determination apparatus of the present invention, a Softmax function may be used as an activation function for an output layer of the convolutional neural network.

(15) According to the signal/noise determination apparatus of the present invention, the signal may be represented as a magnetic dipole moment or a current dipole moment.

(16) The present invention is a signal/noise determination method, with using a signal/noise determination apparatus including a plurality of sensors arranged to measure a signal and a noise, for determining whether components of results of measurement by the sensors are from a signal source or a noise source, the method including: recording a determination model used to determine whether components of results of measurement by the sensors expected with hypothetical signal information and hypothetical noise information are from the signal source or the noise source, the determination model generated by machine learning with the measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and determining whether components of the measurement results are from the signal source or the noise source based on the measurement results and the determination model, wherein the signal information includes the position of the signal source and the signal, and the noise information includes the position of the noise source and the noise.

(17) The present invention is a non-transitory computer-readable medium including a program of instructions for execution by a computer to perform a signal/noise determination process, with using a signal/noise determination apparatus including a plurality of sensors arranged to measure a signal and a noise, for determining whether components of results of measurement by the sensors are from a signal source or a noise source, the signal/noise determination process including: recording a determination model used to determine whether components of results of measurement by the sensors expected with hypothetical signal information and hypothetical noise information are from the signal source or the noise source, the determination model generated by machine learning with the measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and determining whether components of the measurement results are from the signal source or the noise source based on the measurement results and the determination model, wherein the signal information includes the position of the signal source and the signal, and the noise information includes the position of the noise source and the noise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a functional block diagram showing the configuration of a signal/noise determination apparatus 1 according to an embodiment of the present invention;

(2) FIG. 2 is a plan view of a sensor group 12; and

(3) FIGS. 3 (a) and 3 (b) show a positional relationship of the sensor group 12, a signal source region S, and a noise source region N, as a plan view (FIG. 3 (a)) and a front view (FIG. 3 (b)) of the sensor group 12, etc.

DESCRIPTION OF THE PREFERRED EMBODIMENT

(4) A preferred embodiment of the present invention will hereinafter be described with reference to the accompanying drawings.

(5) FIG. 1 is a functional block diagram showing the configuration of a signal/noise determination apparatus 1 according to an embodiment of the present invention. FIG. 2 is a plan view of a sensor group 12. The signal/noise determination apparatus 1 according to the embodiment of the present invention includes a sensor group 12, a determination model recording section 14, and a signal/noise determining section 16.

(6) The sensor group 12 includes multiple sensors 12a. The multiple sensors 12a are arranged to measure a signal and a noise. The signal and the noise are each represented as, for example, a magnetic dipole moment and a current dipole moment. Referring to FIG. 2, the multiple sensors 12a are arranged in a 2-dimensional manner, for example, in an equally spaced manner vertically and horizontally.

(7) FIG. 3 shows a positional relationship of the sensor group 12, a signal source region S, and a noise source region N, as a plan view (FIG. 3 (a)) and a front view (FIG. 3 (b)) of the sensor group 12, etc.

(8) Signal sources S1, S2 are each arranged to output a signal. Noise sources N1, N2 are each arranged to output a noise. It is noted that while there are two signal sources and two noise sources in the example shown in FIG. 3, there may be three or more signal sources with only one or more noise sources required. It is, however, necessary that the total number of signal sources and noise sources be smaller than the

number of the sensors **12a**.

(9) The signal source region **S** is a predetermined single region located almost directly above the sensor group **12**. The signal sources **S1**, **S2** exist within the signal source region **S**. The noise source region **N** is located around the signal source region **S**. The noise sources **N1**, **N2** exist within the noise source region **N**.

(10) The determination model recording section **14** records a determination model used to determine whether components of results of measurement by the sensor group **12** (multiple sensors **12a**) are from the signal sources **S1**, **S2** or the noise sources **N1**, **N2**. A method of generating the determination model will hereinafter be described.

(11) The signal/noise determining section **16** is arranged to determine whether components of the measurement result are from the signal sources **S1**, **S2** or the noise sources **N1**, **N2** based on the measurement results from the sensor group **12** and the determination model recorded in the determination model recording section **14**.

(12) It is noted that the signal/noise determining section **16** is arranged to receive the measurement results from the sensor group **12** and obtain components of the measurement results through independent component analysis or principal component analysis. The signal/noise determining section **16** may also be arranged to obtain components of the measurement results through FFT on the measurement results and further through IFFT on the result of the FFT.

(13) It is noted that the number of components of the measurement results (i.e., the total number of signal sources and noise sources) must have been known in order to perform independent component analysis. It is necessary to estimate the number of components of the measurement results, if has not been known. Here, when eigenvalues (singular values) are calculated from the covariance matrix of the measurement results and arranged in descending order, the number of higher eigenvalues corresponds to the number of components of the measurement results. It is therefore possible for the signal/noise determining section **16** to estimate the number of components of the measurement results by calculating eigenvalues (singular values) from the covariance matrix of the measurement results and obtaining the number of eigenvalues equal to or higher than a predetermined threshold value.

(14) The determination model is arranged to be generated by machine learning (e.g. based on a convolutional neural network) with results of measurement by the sensors **12a** expected with hypothetical signal information (i.e. the position of each signal source and the signal) and hypothetical noise information (i.e. the position of each noise source and the noise) (so-called simulation), the hypothetical signal information, and the hypothetical noise information as training data.

(15) It is noted that upon generation of the determination model, the signal source is assumed to be arranged randomly within the signal source region **S** and the noise source is assumed to be arranged randomly within the noise source region **N**.

(16) Here, if the signal and the noise are each represented as a magnetic dipole moment (vector **m**), the magnetic flux density **B** (function of vector **r**) generated at each of the sensors **12a** is expressed by Biot-Savart's law as in the following formula (1), where $\mu_{\text{sub}0}$ represents the magnetic constant, and the vector **r** is the direction vector from the signal source and the noise source (magnetic dipoles) to the sensor **12a**.

$$(17) \quad B(\vec{r}) = \frac{\mu_0}{4\pi} \left\{ \frac{3(\vec{m} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3} \right\} \quad (1)$$

(18) If the sensor **12a** is a triaxial sensor, the X component B_x , the Y component B_y , and the Z component B_z of the magnetic flux density **B** (function of vector **r**) can be measured and are expressed as in the following respective formulae (2), (3), and (4), where r_x , r_y , and r_z represent, respectively, the X component, the Y component, and the Z component of the vector **r**, and m_x , m_y , and m_z represent, respectively, the X component, the Y component, and the Z component of the vector **m**.

$$(19) \quad B_x = \frac{\mu_0}{4\pi} \left\{ \left(\frac{3r_x^2}{r^5} - \frac{1}{r^3} \right) m_x + \frac{3r_x r_y}{r^5} m_y + \frac{3r_x r_z}{r^5} m_z \right\} \quad (2)$$

$$B_y = \frac{\mu_0}{4\pi} \left\{ \frac{3r_x r_y}{r^5} m_x + \left(\frac{3r_y^2}{r^5} - \frac{1}{r^3} \right) m_y + \frac{3r_y r_z}{r^5} m_z \right\} \quad (3)$$

$$B_z = \frac{1}{4} \left\{ \frac{3r_z r_x}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} m_x + \frac{3r_y r_z}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} m_y + \left(\frac{3r_z^2}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} - \frac{1}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} \right) m_z \right\} \quad (4)$$

(20) With the hypothetical signal information (i.e. the position of the signal source and the signal) and the hypothetical noise information (i.e. the position of the noise source and the noise), the vector r can be obtained based on the positions of the hypothetical signal source and the hypothetical noise source. In addition, the hypothetical signal and the hypothetical noise form the vector m . These vectors r and m can then be substituted into the above formulae (2), (3), and (4) to obtain expected results of measurement by the sensor **12a**. It is here necessary to adjust the average of the magnetic flux density B in the time direction to 0.

(21) Alternatively, if the signal and the noise are each represented as a current dipole moment (vector p), the magnetic flux density B (function of vector r) generated at each of the sensors **12a** is expressed by Biot-Savart's law as in the following formula (5).

$$(22) \quad B(\text{fwdarw.}) = \frac{1}{4} \frac{(\text{fwdarw.} \times \text{fwdarw.})}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} \quad (5)$$

(23) If the sensor **12a** is a triaxial sensor, the X component B_x , the Y component B_y , and the Z component B_z of the magnetic flux density B (function of vector r) can be measured and are expressed as in the following respective formulae (6), (7), and (8), where r_x , r_y , and r_z represent, respectively, the X component, the Y component, and the Z component of the vector r , and p_x , p_y , and p_z represent, respectively, the X component, the Y component, and the Z component of the vector p .

$$(24) \quad B_x = \frac{1}{4} \left(\frac{r_z}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_y - \frac{r_y}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_z \right) \quad (6)$$

$$B_y = \frac{1}{4} \left(\frac{r_x}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_z - \frac{r_z}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_x \right) \quad (7)$$

$$B_z = \frac{1}{4} \left(\frac{r_y}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_x - \frac{r_x}{\text{Math.} \cdot \text{fwdarw.} \cdot \text{Math.}} p_y \right) \quad (8)$$

(25) With the hypothetical signal information (i.e. the position of the signal source and the signal) and the hypothetical noise information (i.e. the position of the noise source and the noise), the vector r can be obtained based on the positions of the hypothetical signal source and the hypothetical noise source. In addition, the hypothetical signal and the hypothetical noise form the vector p . These vectors r and p can then be substituted into the above formulae (6), (7), and (8) to obtain expected results of measurement by the sensor **12a**. It is here necessary to adjust the average of the magnetic flux density B in the time direction to 0.

(26) The magnetic field generated by a coil of radius a , number of turns n , and current I can be expressed in a cylindrical coordinate system as in the following formulae (9), (10), and (11).

$$(27) \quad B_z = \frac{nI}{2} \frac{1}{\sqrt{(a+r)^2 + z^2}} \left\{ K(k) + \frac{a^2 - r^2 - z^2}{(a-r)^2 + z^2} E(k) \right\} \quad (9)$$

$$B_r = \frac{nI}{2} \frac{z}{\sqrt{(a+r)^2 + z^2}} \left\{ -K(k) + \frac{a^2 + r^2 + z^2}{(a-r)^2 + z^2} E(k) \right\} \quad (10) \quad B_\theta = 0 \quad (11)$$

(28) Here, $K(k)$ represents the complete elliptic integral of the first kind and $E(k)$ represents the complete elliptic integral of the second kind, as expressed in the following formulae (12), (13), and (14).

$$(29) \quad K(k) = \int_0^\pi \frac{d}{\sqrt{1 - k^2 \sin^2}} \quad (12) \quad E(k) = \int_0^\pi \sqrt{1 - k^2 \sin^2} \, d \quad (13) \quad k^2 = \frac{4ar}{(a+r)^2 + z^2} \quad (14)$$

(30) With the hypothetical signal information (i.e. the position of the signal source and the signal) and the hypothetical noise information (i.e. the position of the noise source and the noise), the vector r can be obtained based on the positions of the hypothetical signal source and the hypothetical noise source. In addition, the hypothetical signal and the hypothetical noise form the magnetic field generated by a coil of radius a , number of turns n , and current I . The vector r and the radius a , number of turns n , and current I can then be substituted into the above formulae (9), (10), and (11) to obtain expected results of measurement by the sensor **12a**. It is here necessary to adjust the average of the magnetic flux density B in the time direction to 0.

(31) It is noted that in the above-described sensor group **12**, the sensors **12a** capable of measuring triaxial components are arranged in a 2-dimensional manner. It is, however, conceivable that the sensors **12a** may be arranged in a line-shape in a 1-dimensional manner. It is, alternatively, conceivable that the sensors

12a may be arranged sterically in a 3-dimensional manner (as a sensor array). Furthermore, the sensors **12a** may be capable of measuring only a monoaxial component or biaxial components.

(32) When the machine learning method is based on a convolutional neural network, the determination model has an input layer, a convolutional layer, a pooling layer, a fully connected layer, and an output layer. When the measurement results from the sensor **12a** are provided to the input layer, the output from the input layer is provided to the convolutional layer, the output from the convolutional layer is provided to the pooling layer, the output from the pooling layer is provided to the fully connected layer, and the output from the fully connected layer is provided to the output layer. Whether components of the measurement results are from a signal source (output 1) or a noise source (output 0) is output from the output layer.

(33) It is noted that while one convolutional layer and one pooling layer may be provided as described above, two or more convolutional layers and two or more pooling layers may be provided depending on the number of sensors **12a**.

(34) Also, a sigmoid function is used as an activation function for the output layer. However, a Softmax function may be used as an activation function for the output layer. In this case, since the output is multivalued, the distance of the signal source and the noise source from the sensor **12a** can be identified roughly. This improves the accuracy when estimating the positions of the signal sources **S1**, **S2** using the measurement results from the sensor **12a**.

(35) Next will be described an operation according to the embodiment of the present invention.

(36) A determination model is first prepared and recorded in the determination model recording section **14**. The determination model is prepared based on a convolutional neural network. Used as training data are results of measurement by the sensors **12a** expected with hypothetical signal information (i.e. the position of each signal source and the signal) and hypothetical noise information (i.e. the position of each noise source and the noise) (so-called simulation), the hypothetical signal information, and the hypothetical noise information.

(37) The actual signal sources **S1**, **S2** and the actual noise sources **N1**, **N2** (see FIG. 3) are then measured with the sensor group **12**.

(38) The measurement results from the sensors **12a** of the sensor group **12** are provided to the signal/noise determining section **16**. Components of the measurement results are obtained through, for example, independent component analysis. It is further determined whether components of the measurement result are from the signal sources **S1**, **S2** or the noise sources **N1**, **N2** based on the measurement results from the sensor group **12** and the determination model recorded in the determination model recording section **14**.

(39) In accordance with the embodiment of the present invention, when it is determined through machine learning whether components of a measured signal are from the signal sources **S1**, **S2** or the noise sources **N1**, **N2**, used as training data are results of measurement by the sensors **12a** expected with hypothetical signal information (i.e. the position of each signal source and the signal) and hypothetical noise information (i.e. the position of each noise source and the noise) (so-called simulation), the hypothetical signal information, and the hypothetical noise information. Thus, in accordance with the embodiment of the present invention, the training data can be acquired more easily compared to the case where an actual measurement result is used as training data.

(40) Incidentally, the above-described embodiment may be achieved as follows. A computer including a CPU, a hard disk, and a medium (USB memory, CD-ROM, or the like) reading device is caused to read a medium with a program recorded thereon that achieves the above-described components (e.g. the determination model recording section **14** and the signal/noise determining section **16**) and install the program in the hard disk. The above-described features can also be achieved in this manner.

DESCRIPTION OF REFERENCE NUMERAL

(41) **1** Signal/Noise Determination Apparatus **12** Sensor Group **12a** Sensors **14** Determination Model Recording Section **16** Signal/Noise Determining Section **S1**, **S2** Signal Sources **N1**, **N2** Noise Sources **S** Signal Source Region **N** Noise Source Region

Claims

1. A signal/noise determination apparatus, comprising: a plurality of sensors arranged to measure a signal and a noise; a memory that stores a determination model used to determine whether components of expected measurement results of the sensors using hypothetical signal information and hypothetical noise information are from a signal source or a noise source, the determination model generated by machine learning with the expected measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and a processor that determines whether components of actual measurement results are from the signal source or the noise source based on the actual measurement results and the determination model, wherein the hypothetical signal information includes a first position of the signal source and the signal, and the hypothetical noise information includes a second position of the noise source and the noise.
2. The signal/noise determination apparatus according to claim 1, wherein the signal source is assumed to be arranged within a predetermined single region, and the noise source is assumed to be arranged around the region.
3. The signal/noise determination apparatus according to claim 2, wherein the signal source or the noise source is assumed to be arranged randomly.
4. The signal/noise determination apparatus according to claim 1, wherein the components are obtained through independent component analysis on the actual measurement results.
5. The signal/noise determination apparatus according to claim 1, wherein the components are obtained through principal component analysis on the actual measurement results.
6. The signal/noise determination apparatus according to claim 1, wherein the components are obtained through Fast Fourier Transform (FFT) on the actual measurement results and further through Inverse Fast Fourier Transform (IFFT) on a result of the FFT.
7. The signal/noise determination apparatus according to claim 1, wherein the machine learning is based on a convolutional neural network.
8. The signal/noise determination apparatus according to claim 7, wherein a sigmoid function is used to activate an output layer of the convolutional neural network.
9. The signal/noise determination apparatus according to claim 7, wherein a Softmax function is used to activate an output layer of the convolutional neural network.
10. The signal/noise determination apparatus according to claim 1, wherein the signal is represented as a magnetic dipole moment or a current dipole moment.
11. A signal/noise determination method, with using a signal/noise determination apparatus including a plurality of sensors arranged to measure a signal and a noise, for determining whether components of actual measurement results by the sensors are from a signal source or a noise source, the signal/noise determination method comprising: storing, in a memory, a determination model used to determine whether components of expected measurement results of the sensors using hypothetical signal information and hypothetical noise information are from the signal source or the noise source, the determination model generated by machine learning with the expected measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and determining, by a processor, whether the components of the actual measurement results are from the signal source or the noise source based on the actual measurement results and the determination model, wherein the hypothetical signal information includes a first position of the signal source and the signal, and the hypothetical noise information includes a second position of the noise source and the noise.
12. A non-transitory computer-readable medium including a program of instructions for execution by a computer to perform a signal/noise determination process, with using a signal/noise determination apparatus including a plurality of sensors arranged to measure a signal and a noise, for determining whether components of actual measurement results by the sensors are from a signal source or a noise source, the signal/noise determination process comprising: storing, in a memory, a determination model used to determine whether components of expected measurement results of the sensors using hypothetical signal information and hypothetical noise information are from the signal source or the noise source, the determination model generated by machine learning with the expected measurement results, the hypothetical signal information, and the hypothetical noise information as training data; and determining, by a processor, whether the components of the actual measurement results are from the signal source or

the noise source based on the actual measurement results and the determination model, wherein the hypothetical signal information includes a first position of the signal source and the signal, and the hypothetical noise information includes a second position of the noise source and the noise.
